

Sandeep Gautam

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Professional Summary

Geomatics Engineer with strong experience in GIS, remote sensing, and spatial data analysis, working across mapping, hydrological studies, and land use–land cover modeling. Skilled in LiDAR, UAV and GNSS data processing and satellite imagery analysis, with growing expertise in Geospatial Artificial Intelligence (GeoAI) including U-Net and YOLO-based deep learning for automated feature extraction. Experienced in data quality control, spatial modeling, and translating complex geospatial and AI-driven outputs into practical solutions for planning, disaster risk assessment, and environmental management.

Education

Pashchimanchal Campus, Tribhuvan University <i>Bachelor's in Geomatics Engineering (Aggregate 74.81%)</i>	2021 - 2025 <i>Lamachaur, Pokhara</i>
• Relevant Coursework: Fundamentals of Surveying, Fundamentals of GIS, Cartography, Remote Sensing, Photogrammetry, Satellite Positioning, Python Programming	
SOS Hermann Gmeiner School, Gandaki <i>Higher Secondary Education, Science Stream (CGPA 3.58)</i>	2018 - 2020 <i>Rambazar, Pokhara</i>

Experience

Geo 3D Modelling <i>Geomatics Engineer</i>	Apr 2025 – Present <i>Kathmandu, Nepal</i>
• Processed raw LiDAR and UAV data to classified point clouds, DEM's, 3D Models and Contours. • Processed bathymetry data to determine lake morphometry and for sedimentation studies. • Utilized satellite imageries for LULC Classification, vegetation monitoring, temperature monitoring and urban monitoring. • Utilized SAR images for flood monitoring and deformation monitoring. • Digitized groundwater irrigation command areas for water planning. • Performed automation for Geospatial analysis using Python and ModelBuilder tools. • Performed QA/QC, spatial validation, attribute correction, and map digitization. • Supported field teams through GNSS/UAV/LiDAR/Bathymetry/TotalStation surveys and data verification.	
National Geographic Society+ The Nature Conservancy by Extern <i>Freshwater and Community Conservation Remote Extern</i>	July 2025 – Sep 2025
• Conducted landscape and gap analysis to assess freshwater challenges in Phewa Lake. • Engaged with community stakeholders and local leaders for participatory mapping and data collection. • Reviewed research papers and conducted interviews for evidence based assessment. • Developed an interactive Story Map dashboard to communicate findings to policymakers and the public	
Freelance GIS Analyst / Consultant <i>Remote (US & Europe Clients)</i>	2023–Present
• Updated and standardized 7,000+ address points using parcel, right-of-way, and satellite imagery, delivering topologically clean GeoPackage datasets. • Digitized 1 million+ feet of roadway landbase features (Edge of Pavement, driveways, sidewalks) with 1-ft accuracy using high-resolution imagery. • Performed spatial validation, QA/QC, and attribute normalization to ensure integration-ready geospatial datasets.	

Projects

Deep Learning for Automated Geospatial Feature Extraction

2026

Personal Project

- Developed and optimized a **U-Net** architecture for semantic segmentation of building footprints and cotton crops from multi-spectral satellite imagery, achieving validation **IoU of 0.88 (buildings) and 0.93 (crops)**.
- Fine-tuned a pretrained **YOLOv8** object detection model for automated building detection, achieving **96.1% mAP@50** and **80% mAP@50-95** with 95% precision and 93% recall.
- Implemented a hybrid **BCE + Dice Loss** to mitigate severe class imbalance and improve boundary delineation accuracy in segmentation tasks.
- Built an end-to-end geospatial deep learning pipeline using **GDAL, NumPy, GeoPandas, Rasterio, TensorFlow/Keras, and PyTorch (Ultralytics)**, converting model outputs into GIS-ready georeferenced vector layers for spatial analysis and mapping.

[GitHub Repository](#)

Spatio-temporal Assessment and Prediction of Urban Heat Islands in Kathmandu Valley

2024–2025

Final Year Project

- Processed multi-temporal Landsat imagery (2008–2023) to generate LULC, NDVI, NDBI, LST, and UHI maps.
- Applied CA–Markov modeling and **Random Forest Regression** to predict 2028 LULC and land surface temperature.
- Analyzed statistical relationships between NDVI/NDBI and LST to quantify drivers of urban heat island formation.
- Visualized predictive outputs using ArcGIS and Python for technical reporting and urban planning recommendations.

[Poster Link](#)

WGS/UTM to MUTM Coordinate Transformation Tool

2026

Web-Based Geospatial Tool

- Developed an automated web-based tool for converting standard UTM spatial data to MUTM coordinate systems.
- Implemented official Nepal Survey Department parameters to ensure geodetic accuracy for regional engineering projects.
- Streamlined legacy data integration for civil engineers and urban planners by removing complex GIS processing requirements.
- Validated tool output against industry-standard GIS software to guarantee professional-grade reliability.

[Tool Link](#)

A StoryMap on Freshwater Challenges and Community Conservation of Phewa Lake

2025

Externship Project

Phewa Lake, Kaski, Nepal

- Integrated primary and secondary geospatial and socio-environmental datasets related to freshwater challenges and community impacts.
- Designed and published an ArcGIS StoryMap to communicate findings, narratives, and visuals to local stakeholders and policymakers.

[Story Map Link](#)

Bathymetry Survey and Sedimentation Study of Phewa, Begnas and Rupa Lake

2025-2026

Geo 3D Modelling

Pokhara, Nepal

- Performed field operations using single beam RTK echo-sounder.
- Generated DEM's and Contours to determine morphometric characteristics of the lakes.
- Reviewed past survey informations from secondary sources and utilized it to perform a in-depth sedimentation study and research.

Hydrological Study and Flood Risk Assessment – Chandrouta Solar Site

2025

Geo3D Modelling

Kapilvastu, Nepal

- Conducted hydrological and flood risk assessment for a 94.38 sq. km watershed using GIS, remote sensing, UAV-derived DEM, and HEC-RAS modeling.

- Processed drone survey data with GCPs to generate high-resolution DEM and orthomosaic for watershed delineation and hydraulic analysis.
- Performed land use-land cover mapping using Sentinel-2 imagery and field verification to support runoff and flood modeling.
- Analyzed 25 years of rainfall data from DHM and prepared flood inundation maps for 2, 5, 10, 25, and 50-year return periods.
- Prepared flood risk maps and technical recommendations for drainage planning and raised infrastructure for solar farm development.

Publications & Research

Mitigating Class Imbalance and Negative Learning Through Object Centric Patch Sampling in Remote Sensing

Under Review: MDPI Remote Sensing

Spatio-Temporal Assessment of Urban Heat Islands and Future Prediction: A Case Study of Kathmandu Valley using Optical Remote Sensing

Under Review: Journal of Land Management and Geomatics Education

Technical Skills

Geospatial & Remote Sensing Tools: ArcGIS, QGIS, Google Earth Engine, Global Mapper Pro, Pix4D Mapper, LiGeoreference, LiDAR360

AI & Deep Learning (GeoAI): U-Net, YOLOv8 (Ultralytics), Convolutional Neural Networks (CNNs), Object Detection, Semantic Segmentation, TensorFlow, Keras, PyTorch, Scikit-learn

Programming & Geospatial Data: Python (GDAL, GeoPandas, Rasterio, NumPy, Pandas), basic SQL

Hydrological & Spatial Modeling: HEC-RAS, Watershed Delineation, LULC Classification, Flood Risk Modeling

Field & Data Collection: GPS/DGPS, UAV Surveying, KoboToolbox

Web & Visualization: Leaflet.js, OpenStreetMap editors, ArcGIS StoryMaps, Python visualization

Office & Communication: MS Word, Excel, PowerPoint

Trainings and Workshops

Climbing for Climate Geospatial School (CLIGEOS-2024): Applications of GIS, Remote Sensing, AI, and GeoBigData in climate change; field mapping using drones and open-source GIS tools.

Workshop on AI-Supported Landslide Detection and Susceptibility Analysis: Remote sensing foundations, Sentinel-1 processing, and AI-based susceptibility mapping.

7-Day WebGIS Training: Basics of WebGIS workflows and tools.

Additional Information

Soft Skills: Team collaboration in multi-disciplinary settings, critical thinking under tight timelines, community engagement, technical documentation, report writing

Languages: Nepali (Fluent), English (Fluent), Hindi (Conversational)

References

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